of the pandemic, Linux experienced a surge in desktop usage. However, as the world adjusted to the new normal, its popularity dipped below 1.5%. Speculation arose that Linux was primarily used in homes rather than workplaces. Yet, in the current landscape, Linux has made a resounding comeback, doubling its usage compared to three years ago and surpassing the 3% milestone.

Notably, this is the first time Linux has achieved a market share above 3% on desktops. Recent statistics from StatCounter indicate Linux reached 3.07% in global market share in June, with Chrome OS maintaining a separate section at 4.13%. macOS claimed 21.32%, while the long-standing market leader, Windows, still reigns supreme with 68.23%, although this represents a significant decline from its peak of nearly 80% in July 2022.

While StatCounter's data provides insights, the reasons behind Linux's breakthrough remain uncertain. Factors such as the forthcoming release of Windows 11, which imposes hardware restrictions, may have led some users to explore Linux as an alternative. Additionally, Valve's introduction of the Steam Deck, a handheld gaming device that supports Linux, could have contributed to more game developers embracing the platform.

The key question now is whether this achievement represents a temporary spike or a sustained trend. Only time will reveal the true significance of this milestone and shed light on Linux's future trajectory in the highly competitive desktop operating system landscape.

Linux kernel 6.3 reaches end of life; the StackRot vulnerability is fixed in version 6.4

Linux kernel 6.3 has reached its end of life and will no longer be supported by bug and security fixes. This announcement comes amidst the discovery of a critical vulnerability, dubbed StackRot (CVE-2023-3269), which affects Linux kernel versions 6.1 through 6.4. Attackers can exploit this vulnerability to escalate privileges on compromised systems.

Security researcher Ruihan Li of Peking University in China uncovered the vulnerability, describing it as a pervasive issue that affects almost all Linux kernel configurations. Li emphasised that triggering the vulnerability requires minimal capabilities, making it a concerning threat to Linux users.

In response to the discovery, a dedicated team, led by Linux creator Linus Torvalds, worked tirelessly for two weeks to develop a set of patches addressing the vulnerability. The fix was merged into Linus' tree during the merge window for Linux kernel 6.5 on June 28th.

StackRot revolves around the Linux kernel's handling of stack expansion—a mechanism that automatically increases the stack memory of a running process. Li explained that the flaw arises due to a memory management function in the Linux kernel's data structure for managing virtual memory spaces. This flaw results in use-after-free-by-RCU (UAFBR) issues, combining the use-after-free vulnerability with the Read-Copy-Update (RCU) mechanism for synchronising shared data.

Use-after-free vulnerabilities pose a serious threat as they allow attackers to insert arbitrary code into freed, yet still used, memory space. Exploiting such vulnerabilities is challenging due to a delay in memory deallocation caused by RCU callbacks. However, StackRot represents a first-of-its-kind successful exploitation of a UAFBR bug.

To address the vulnerability, the Linux kernel team, spearheaded by Torvalds,

PolyLM, an open source multilingual LLM, unveiled

Researchers from DAMO Academy and Alibaba Group have introduced PolyLM, an open source multilingual large language model (LLM) designed to address the limitations of existing models. Available in two model sizes, 1.7B and 13B, PolyLM offers advanced capabilities in understanding, reasoning, and generating text across multiple languages.

PolyLM excels in Spanish, Russian, Arabic, Japanese, Korean, Thai, Indonesian, and Chinese, complementing existing models. Its training strategy facilitates knowledge transfer from English to other languages, enhancing its multilingual performance. To improve understanding of multilingual instructions, PolyLM utilises the MULTIALPACA data set, which provides high-quality multilingual instruction data.

The researchers utilised a massive data set of 640B tokens from sources like Wikipedia, mC4, and CC-100 to train PolyLM. They employed a curricular learning technique, gradually increasing the focus on low-resource languages while initially concentrating on English. This approach ensured the transfer of general knowledge across languages.

PolyLM's evaluation involved a benchmark comprising multilingual tasks such as question answering, language understanding, text generation, and cross-lingual machine translation. The experiments showcased PolyLM's superior performance in non-English languages compared to existing models of similar size. The model's multilingual capabilities were further enhanced using multilingual instruction data.

With PolyLM's introduction, the AI community now has access to a powerful multilingual LLM. Its proficiency in major non-English languages and advanced training techniques makes it a significant milestone in the field of natural language processing.